

1. NAME OF THE MEDICINAL PRODUCT

VELCADE® 3.5 mg for injection (sterile lyophilized powder)

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

VELCADE® 3.5 mg for injection (sterile lyophilized powder) is an antineoplastic agent available for intravenous injection (IV) or subcutaneous (SC) use. Each single use vial contains:

- 3.5 mg of bortezomib as a sterile lyophilized powder. Inactive ingredient: 35 mg mannitol, USP/EP (IV or SC use).

For excipients, see section 6.1.

3. PHARMACEUTICAL FORM

VELCADE® 3.5 mg for injection (sterile lyophilized powder) is supplied as individually cartoned 10 mL vials containing 3.5 mg of bortezomib as a white to off-white cake or powder.

- 3.5 mg single use vial

4. CLINICAL PARTICULARS

4.1 Therapeutic Indications

VELCADE is indicated for the treatment of patients with multiple myeloma.

VELCADE is indicated for the treatment of patients with mantle cell lymphoma.

4.2 Posology and Method of Administration

Important Dosing Guidelines

VELCADE is for intravenous or subcutaneous use only. Do not administer VELCADE by any other route.

Because each route of administration has a different reconstituted concentration, use caution when calculating the volume to be administered.

The recommended starting dose of VELCADE is 1.3 mg/m². VELCADE is administered intravenously at a concentration of 1 mg/ml, or subcutaneously at a concentration of 2.5 mg/ml.

VELCADE retreatment may be considered for patients with multiple myeloma who had previously responded to treatment with VELCADE and who have relapsed at least six months after completing prior VELCADE treatment. Treatment may be started at the last tolerated dose.

When administered intravenously, administer VELCADE as a 3 to 5 second bolus intravenous injection.

Dosage in Previously Untreated Multiple Myeloma

VELCADE is administered in combination with oral melphalan and oral prednisone for 9, six week treatment cycles as shown in **Table 1**. In Cycles 1 to 4, VELCADE is administered twice weekly (Days 1, 4, 8, 11, 22, 25, 29 and 32). In Cycles 5 to 9, VELCADE is administered once weekly (Days 1, 8, 22 and 29). At least 72 hours should elapse between consecutive doses of VELCADE.

Table 1: Dosage Regimen for Patients with Previously Untreated Multiple Myeloma

Twice Weekly VELCADE (Cycles 1 to 4)												
Week	1				2		3	4		5		6
VELCADE (1.3 mg/m ²)	Day 1	--	--	Day 4	Day 8	Day 11	rest period	Day 22	Day 25	Day 29	Day 32	rest period
Melphalan (9 mg/m ²) Prednisone (60 mg/m ²)	Day 1	Day 2	Day 3	Day 4	--	--	rest period	--	--	--	--	rest period
Once Weekly VELCADE (Cycles 5 to 9 when used in combination with Melphalan and Prednisone)												
Week	1				2		3	4		5		6
VELCADE (1.3 mg/m ²)	Day 1	--	--		Day 8		rest period	Day 22		Day 29		rest period
Melphalan (9 mg/m ²) Prednisone (60 mg/m ²)	Day 1	Day 2	Day 3	Day 4	--	--	rest period	--	--	--	--	rest period

Dose Modification Guidelines for VELCADE when given in Combination with Melphalan and Prednisone

Prior to initiating any cycle of therapy with VELCADE in combination with melphalan and prednisone:

- Platelet count should be at least $70 \times 10^9/L$ and the absolute neutrophil count (ANC) should be at least $1 \times 10^9/L$
- Nonhematological toxicities should have resolved to Grade 1 or baseline

Table 2: Dose Modifications during Cycles of Combination VELCADE, Melphalan and Prednisone Therapy

Toxicity	Dose Modification or Delay
Hematological toxicity during a cycle: If prolonged Grade 4 neutropenia or thrombocytopenia, or thrombocytopenia with bleeding is observed in the previous cycle	Consider reduction of the melphalan dose by 25% in the next cycle
If platelet count is not above $30 \times 10^9/L$ or ANC is not above $0.75 \times 10^9/L$ on a VELCADE dosing day (other than day 1)	Withhold VELCADE dose

If several VELCADE doses in consecutive cycles are withheld due to toxicity	Reduce VELCADE dose by 1 dose level (from 1.3 mg/m ² to 1 mg/m ² , or from 1 mg/m ² to 0.7 mg/m ²)
Grade 3 or higher non-hematological toxicities	Withhold VELCADE therapy until symptoms of toxicity have resolved to Grade 1 or baseline. Then, VELCADE may be reinitiated with one dose level reduction (from 1.3 mg/m ² to 1 mg/m ² , or from 1 mg/m ² to 0.7 mg/m ²). For VELCADE-related neuropathic pain and/or peripheral neuropathy, hold or modify VELCADE as outlined in Table 6 .

For information concerning melphalan and prednisone, see manufacturer's prescribing information.

Dose modifications guidelines for peripheral neuropathy are provided.

Posology for previously untreated multiple myeloma patients eligible for haematopoietic stem cell transplantation (induction therapy)

Combination therapy with dexamethasone

VELCADE 3.5 mg powder for solution for injection is administered via intravenous or subcutaneous injection at the recommended dose of 1.3 mg/m² body surface area twice weekly for two weeks on days 1, 4, 8, and 11 in a 21-day treatment cycle. This 3-week period is considered a treatment cycle. At least 72 hours should elapse between consecutive doses of VELCADE.

Dexamethasone is administered orally at 40 mg on days 1, 2, 3, 4, 8, 9, 10 and 11 of the VELCADE treatment cycle.

Four treatment cycles of this combination therapy are administered.

Combination therapy with dexamethasone and thalidomide

VELCADE 3.5 mg powder for solution for injection is administered via intravenous or subcutaneous injection at the recommended dose of 1.3 mg/m² body surface area twice weekly for two weeks on days 1, 4, 8, and 11 in a 28-day treatment cycle. This 4-week period is considered a treatment cycle. At least 72 hours should elapse between consecutive doses of VELCADE.

Dexamethasone is administered orally at 40 mg on days 1, 2, 3, 4, 8, 9, 10 and 11 of the VELCADE treatment cycle.

Thalidomide is administered orally at 50 mg daily on days 1-14 and if tolerated the dose is increased to 100 mg on days 15-28, and thereafter may be further increased to 200 mg daily from cycle 2 (see **Table 3**).

Four treatment cycles of this combination are administered. It is recommended that patients with at least partial response receive 2 additional cycles.

Table 3: Posology for VELCADE combination therapy for patients with previously untreated multiple myeloma eligible for haematopoietic stem cell transplantation

Vc+ Dx	Cycles 1 to 4			
	Week	1	2	3

	Vc (1.3 mg/m ²)	Day 1, 4	Day 8, 11	Rest Period
	Dx 40 mg	Day 1, 2, 3, 4	Day 8, 9, 10, 11	-
Vc+Dx+T	Cycle 1			
	Week	1	2	3
	Vc (1.3 mg/m ²)	Day 1, 4	Day 8, 11	Rest Period
	T 50 mg	Daily	Daily	-
	T 100 mg ^a	-	-	Daily
	Dx 40 mg	Day 1, 2, 3, 4	Day 8, 9, 10, 11	-
	Cycles 2 to 4^b			
	Vc (1.3 mg/m ²)	Day 1, 4	Day 8, 11	Rest Period
	T 200 mg ^a	Daily	Daily	Daily
	Dx 40 mg	Day 1, 2, 3, 4	Day 8, 9, 10, 11	-

Vc=VELCADE; Dx=dexamethasone; T=thalidomide

^A Thalidomide dose is increased to 100 mg from week 3 of Cycle 1 only if 50 mg is tolerated and to 200 mg from cycle 2 onwards if 100 mg is tolerated.

^B Up to 6 cycles may be given to patients who achieve at least a partial response after 4 cycles

Dosage adjustments for transplant eligible patients

For VELCADE dosage adjustments, as described under “**Dosage and Dose Modifications for Relapsed Multiple Myeloma and Relapsed Mantle Cell Lymphoma**” and “**Dose Modifications for Peripheral Neuropathy**” should be followed.

In addition, when VELCADE is given in combination with other chemotherapeutic medicinal products, appropriate dose reductions for these products should be considered in the event of toxicities according to the recommendations in the local package inserts.

Dosage in Previously Untreated Mantle Cell Lymphoma

VELCADE (1.3 mg/m²) is administered intravenously in combination with intravenous rituximab, cyclophosphamide, doxorubicin and oral prednisone (VcR-CAP) for 6, three week treatment cycles as shown in **Table 4**. VELCADE is administered first followed by rituximab. VELCADE is administered twice weekly for two weeks (Days 1, 4, 8, and 11) followed by a ten day rest period on Days 12 to 21. For patients with a response first documented at Cycle 6, two additional VcR-CAP cycles are recommended. At least 72 hours should elapse between consecutive doses of VELCADE.

Table 4: Dosage Regimen for Patients with Previously Untreated Mantle Cell Lymphoma

Twice Weekly VELCADE (6, Three Week Cycles)*								
Week	1				2		3	
VELCADE (1.3 mg/m ²)	Day 1	--	--	Day 4	--	Day 8	Day 11	rest period
Rituximab (375 mg/m ²) Cyclophosphamide (750 mg/m ²)	Day 1	--	--			--	--	rest period

Doxorubicin (50 mg/m ²)								
Prednisone (100 mg/m ²)	Day 1	Day 2	Day 3	Day 4	Day 5	--	--	rest period

* Dosing may continue for 2 more cycles (for a total of 8 cycles) if response is first seen at cycle 6.

Dose Modification Guidelines for VELCADE When Given in Combination with Rituximab, Cyclophosphamide, Doxorubicin and Prednisone

Prior to the first day of each cycle (other than Cycle 1):

- Platelet count should be at least $100 \times 10^9/L$ and absolute neutrophil count (ANC) should be at least $1.5 \times 10^9/L$
- Hemoglobin should be at least 8 g/dL (at least 4.96 mmol/L)
- Non-hematologic toxicity should have recovered to Grade 1 or baseline

Interrupt VELCADE treatment at the onset of any Grade 3 hematologic or nonhematological toxicities, excluding neuropathy [see **Table 6** and **Special warnings and special precautions for use**]. For dose adjustments, see **Table 5** below.

Table 5: Dose Modifications on Days 4, 8, and 11 during Cycles of Combination VELCADE, Rituximab, Cyclophosphamide, Doxorubicin and Prednisone Therapy

Toxicity	Dose modification or delay
Hematological toxicity • Grade 3 or higher neutropenia, or a platelet count not at or above $25 \times 10^9/L$	Withhold VELCADE therapy for up to 2 weeks until the patient has an ANC at or above $0.75 \times 10^9/L$ and a platelet count at or above $25 \times 10^9/L$. • If, after VELCADE has been withheld, the toxicity does not resolve, discontinue VELCADE. • If toxicity resolves such that the patient has an ANC at or above $0.75 \times 10^9/L$ and a platelet count at or above $25 \times 10^9/L$, VELCADE dose should be reduced by 1 dose level (from 1.3 mg/m^2 to 1 mg/m^2, or from 1 mg/m^2 to 0.7 mg/m^2).
Grade 3 or higher nonhematological toxicities	Withhold VELCADE therapy until symptoms of the toxicity have resolved to Grade 2 or better. Then, VELCADE may be reinitiated with one dose level reduction (from 1.3 mg/m^2 to 1 mg/m^2 , or from 1 mg/m^2 to 0.7 mg/m^2). For VELCADE-related neuropathic pain and/or peripheral neuropathy, hold or modify VELCADE as outlined in Table 6 .

For information concerning rituximab, cyclophosphamide, doxorubicin and prednisone, see manufacturer's prescribing information.

Dosage and Dose Modifications for Relapsed Multiple Myeloma and Relapsed Mantle Cell Lymphoma

VELCADE (1.3 mg/m²/dose) is administered twice weekly for two weeks (Days 1, 4, 8, and 11) followed by a ten day rest period (Days 12 to 21). For extended therapy of more than eight cycles, VELCADE may be administered on the standard schedule or, for relapsed multiple myeloma, on a maintenance schedule of once weekly for four weeks (Days 1, 8, 15, and 22) followed by a 13 day rest period (Days 23 to 35). At least 72 hours should elapse between consecutive doses of VELCADE.

Patients with multiple myeloma who have previously responded to treatment with VELCADE (either alone or in combination) and who have relapsed at least six months after their prior VELCADE therapy may be started on VELCADE at the last tolerated dose. Retreated patients are administered VELCADE twice weekly (Days 1, 4, 8, and 11) every three weeks for a maximum of eight cycles. At least 72 hours should elapse between consecutive doses of VELCADE. VELCADE may be administered either as a single agent or in combination with dexamethasone.

VELCADE therapy should be withheld at the onset of any Grade 3 nonhematological or Grade 4 hematological toxicities excluding neuropathy as discussed below (**see Special warnings and special precautions for use**). Once the symptoms of the toxicity have resolved, VELCADE therapy may be reinitiated at a 25% reduced dose (1.3 mg/m²/dose reduced to 1 mg/m²/dose; 1 mg/m²/dose reduced to 0.7 mg/m²/dose).

For dose modifications guidelines for peripheral neuropathy, see Dose modifications for peripheral neuropathy below.

Dose Modifications for Peripheral Neuropathy

Starting VELCADE subcutaneously may be considered for patients with pre-existing or at high risk of peripheral neuropathy. Patients with pre-existing severe neuropathy should be treated with VELCADE only after careful risk-benefit assessment.

Patients experiencing new or worsening peripheral neuropathy during VELCADE therapy may require a decrease in the dose and/or a less dose-intense schedule.

For dose or schedule modification guidelines for patients who experience VELCADE-related neuropathic pain and/or peripheral neuropathy see **Table 6**.

Table 6: Recommended Dose Modification for VELCADE-related Neuropathic Pain and/or Peripheral Sensory or Motor Neuropathy

Severity of Peripheral Neuropathy Signs and Symptoms*	Modification of Dose and Regimen
Grade 1 (asymptomatic; loss of deep tendon reflexes or paresthesia) without pain or loss of function	No action

Grade 1 with pain or Grade 2 (moderate symptoms; limiting instrumental Activities of Daily Living (ADL)**)	Reduce VELCADE to 1 mg/m ²
Grade 2 with pain or Grade 3 (severe symptoms; limiting self care ADL***)	Withhold VELCADE therapy until toxicity resolves. When toxicity resolves reinstitute with a reduced dose of VELCADE at 0.7 mg/m ² once per week.
Grade 4 (life- threatening consequences; urgent intervention indicated)	Discontinue VELCADE

*Grading based on NCI Common Toxicity Criteria CTCAE v4.0.

**Instrumental ADL: refers to preparing meals, shopping for groceries or clothes, using telephone, managing money etc.

***Self care ADL: refers to bathing, dressing and undressing, feeding self, using the toilet, taking medications, and not bedridden

Special Patient Populations

Impaired Renal Function

The pharmacokinetics of VELCADE are not influenced by the degree of renal impairment. Therefore, dosing adjustments of VELCADE are not necessary for patients with renal insufficiency. Since dialysis may reduce VELCADE concentrations, the drug should be administered after the dialysis procedure (see **Pharmacokinetic Properties, 5.2**).

Impaired Hepatic Function

Patients with mild hepatic impairment do not require a starting dose adjustment and should be treated per the recommended VELCADE dose. For patients with moderate or severe hepatic impairment, see **Table 7** below, (also, see **Pharmacokinetic Properties, 5.2**):

Table 7: Recommended Starting Dose Modification for VELCADE in Patients with Hepatic Impairment

	Bilirubin Level	SGOT (AST) Levels	Modification of Starting Dose in Multiple Myeloma and Relapsed Mantle Cell Lymphoma (1.3 mg/m ² twice weekly)
Mild	≤ 1.0x ULN	> ULN	None
	> 1.0x–1.5x ULN	Any	None
Moderate	> 1.5x–3x ULN	Any	Reduce VELCADE to 0.7 mg/m ² in the first cycle. Consider dose escalation to 1.0 mg/m ² or further dose reduction to 0.5 mg/m ² in subsequent cycles based on patient tolerability.
Severe	> 3x ULN	Any	

Abbreviations: SGOT = serum glutamic oxaloacetic transaminase; AST = aspartate aminotransferase; ULN = upper limit of the normal range.

Method of Administration

VELCADE is administered intravenously or subcutaneously. When administered intravenously, VELCADE is administered as a 3-5 second bolus intravenous injection through a peripheral or central intravenous catheter followed by a flush with 0.9% sodium chloride solution for injection. For subcutaneous administration, the reconstituted solution is injected into the thighs (right or left) or abdomen (right or left). Injection sites should be rotated for successive injections.

If local injection site reactions occur following VELCADE injection subcutaneously, a less concentrated VELCADE solution (1 mg/ml instead of 2.5 mg/ml) may be administered subcutaneously, or changed to IV injection.

4.3 Contraindications

VELCADE is contraindicated in patients with hypersensitivity to bortezomib, boron or mannitol.

4.4 Special Warnings and Special Precautions for Use

VELCADE should be administered under the supervision of a physician experienced in the use of antineoplastic therapy.

There have been fatal cases of inadvertent intrathecal administration of VELCADE. VELCADE is for IV and subcutaneous use only. **DO NOT ADMINISTER VELCADE INTRATHECALLY.**

Overall, the safety profile of patients treated with VELCADE in monotherapy was similar to that observed in patients treated with VELCADE in combination with melphalan and prednisone.

Peripheral Neuropathy

VELCADE treatment causes a peripheral neuropathy (PN) that is predominantly sensory. However, cases of severe motor neuropathy with or without sensory peripheral neuropathy have been reported.

Patients with pre-existing symptoms (numbness, pain or a burning feeling in the feet or hands) and/or signs of peripheral neuropathy may experience worsening peripheral neuropathy (including $\geq$ Grade 3) during treatment with VELCADE. Patients should be monitored for symptoms of neuropathy, such as a burning sensation, hyperesthesia, hypoesthesia, paresthesia, discomfort, neuropathic pain or weakness. In the Phase 3 study comparing VELCADE IV vs SC the incidence of Grade ≥ 2 peripheral neuropathy events was 24% for SC and 41% for IV ($p=0.0124$). Grade ≥ 3 peripheral neuropathy occurred in 6% of subjects in the SC treatment group, compared with 16% in the IV treatment group ($p=0.0264$) (**Table 11**). Therefore, patients with pre-existing PN or at high risk of peripheral neuropathy may benefit from starting VELCADE subcutaneously.

Patients experiencing new or worsening peripheral neuropathy may require a change in dose schedule or route of administration to SC (see **Posology and Method of Administration, 4.2**). Following dose adjustments, improvement in or resolution of peripheral neuropathy was reported in 51% of patients with $\geq$ Grade 2 peripheral neuropathy in the single agent phase 3 multiple myeloma study of VELCADE vs dexamethasone. Improvement in or resolution of peripheral neuropathy was reported in 73% of patients who discontinued due to Grade 2 neuropathy or who had $\geq$ Grade 3 peripheral neuropathy in the phase 2 multiple myeloma studies.

The long-term outcome of peripheral neuropathy has not been studied in mantle cell lymphoma.

Hypotension

In phase 2 and 3 single agent multiple myeloma studies, the incidence of hypotension (postural, orthostatic, and Hypotension Not Otherwise Specified) was 11% to 12%. These events are observed throughout therapy. Caution should be used when treating patients with a history of syncope, patients receiving medications known to be associated with hypotension, and patients who are dehydrated. Management of orthostatic/postural hypotension may include adjustment of antihypertensive medications, hydration, or administration of mineralocorticoids and/or sympathomimetics (see **Undesirable Effects, 4.8**).

Cardiac Disorders

Acute development or exacerbation of congestive heart failure, and/or new onset of decreased left ventricular ejection fraction has been reported, including reports in patients with few or no risk factors for decreased left ventricular ejection fraction. Patients with risk factors for, or existing heart disease should be closely monitored. In the single agent phase 3 multiple myeloma study of VELCADE vs dexamethasone, the incidence of any treatment-emergent cardiac disorder was 15% and 13% respectively. The incidence of heart failure events (acute pulmonary edema, cardiac failure, congestive cardiac failure, cardiogenic shock, pulmonary edema) was similar in the VELCADE and dexamethasone groups, 5% and 4%, respectively.

There have been isolated cases of QT-interval prolongation in clinical studies; causality has not been established.

Hepatic Events

Rare cases of acute liver failure have been reported in patients receiving multiple concomitant medications and with serious underlying medical conditions. Other reported hepatic events include increases in liver enzymes, hyperbilirubinemia, and hepatitis. Such changes may be reversible upon discontinuation of VELCADE. There is limited re-challenge information in these patients.

Pulmonary Disorders

There have been rare reports of acute diffuse infiltrative pulmonary disease of unknown etiology such as pneumonitis, interstitial pneumonia, lung infiltration and Acute Respiratory Distress Syndrome (ARDS) in patients receiving VELCADE. Some of these events have been fatal. A higher proportion of these events have been reported in Japan. In the event of new or worsening pulmonary symptoms, a prompt diagnostic evaluation should be performed and patients treated appropriately.

In a clinical trial, two patients given high-dose cytarabine (2g/m² per day) by continuous infusion with daunorubicin and VELCADE for relapsed acute myelogenous leukemia died of ARDS early in the course of therapy.

Laboratory Tests

Complete blood counts (CBC) should be frequently monitored throughout treatment with VELCADE.

Thrombocytopenia/Neutropenia

VELCADE is associated with thrombocytopenia and neutropenia (see **Undesirable Effects, 4.8**). Platelets were lowest at Day 11 of each cycle of VELCADE treatment and typically recovered to baseline by the next cycle. The cyclical pattern of platelet count decrease and recovery remain consistent in the studies of multiple myeloma and mantle cell lymphoma, with no evidence of cumulative thrombocytopenia or neutropenia in any of the regimens studied.

Platelet counts should be monitored prior to each dose of VELCADE. VELCADE therapy should be held when the platelet count is $<25,000/\mu\text{L}$ (see **Posology and Method of Administration, 4.2** and **Undesirable Effects, 4.8**). There have been reports of gastrointestinal and intracerebral hemorrhage in association with VELCADE. Transfusion and supportive care may be considered.

In the single-agent multiple myeloma study of VELCADE vs dexamethasone, the mean platelet count nadir measured was approximately 40% of baseline. The severity of thrombocytopenia related to pretreatment platelet count is shown in **Table 8**. The incidence of significant bleeding events ($\geq$ Grade 3) was similar on both the VELCADE (4%) and dexamethasone (5%) arms.

Table 8: Severity of Thrombocytopenia Related to Pretreatment Platelet Count in the Single Agent Phase 3 Multiple Myeloma Study of VELCADE vs Dexamethasone

Pretreatment Platelet Count^a	Number of Patients (N = 331)^b	Number (%) of Patients with Platelet Count $< 10,000/\mu\text{L}$	Number (%) of Patients with Platelet Count 10,000-25,000/μL
$\geq 75,000/\mu\text{L}$	309	8 (3%)	36 (12%)
$\geq 50,000/\mu\text{L} - < 75,000/\mu\text{L}$	14	2 (14%)	11 (79%)
$\geq 10,000/\mu\text{L} - < 50,000/\mu\text{L}$	7	1 (14%)	5 (71%)

^a A baseline platelet count of $50,000/\mu\text{L}$ was required for study eligibility.

^b Data were missing at baseline for 1 patient.

In the combination study of VELCADE with rituximab, cyclophosphamide, doxorubicin and prednisone (VcR-CAP) in previously untreated mantle cell lymphoma patients, the incidence of thrombocytopenia adverse events ($\geq$ Grade 4) was 32% versus 2% for the rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone (R-CHOP) arm. The incidence of bleeding adverse events ($\geq$ Grade 3) was 1.7% (4 patients) in the VcR-CAP arm and was 1.2% (3 patients) in the R-CHOP arm.

There were no deaths due to bleeding events in either arm. There were no CNS bleeding events in the VcR-CAP arm; there was 1 bleeding event in the R-CHOP arm. Platelet transfusions were given to 23% of the patients in the VcR-CAP arm and 3% of the patients in the R-CHOP arm.

The incidence of neutropenia ($\geq$ Grade 4) was 70% in the VcR-CAP arm and was 52% in the R-CHOP arm. The incidence of febrile neutropenia ($\geq$ Grade 4) was 5% in the VcR-CAP arm and was 6% in the R-CHOP arm. Colony-stimulating factor support was provided at a rate of 78% in the VcR-CAP arm and 61% in the R-CHOP arm.

Gastrointestinal Adverse Events

VELCADE treatment can cause nausea, diarrhea, constipation, and vomiting (see **Undesirable Effects, 4.8**) sometimes requiring use of antiemetics and antidiarrheal medications. Fluid and electrolyte replacement should be administered to prevent dehydration. Since patients receiving VELCADE therapy may experience vomiting and/or diarrhea, patients should be advised regarding appropriate measures to avoid dehydration. Patients should be instructed to seek medical advice if they experience symptoms of dizziness, light headedness or fainting spells.

Tumor Lysis Syndrome

Because VELCADE is a cytotoxic agent and can rapidly kill malignant cells the complications of tumor lysis syndrome may occur. The patients at risk of tumor lysis syndrome are those with high tumor burden prior to treatment. These patients should be monitored closely and appropriate precautions taken.

Patients with Hepatic Impairment

Bortezomib is metabolized by liver enzymes. Bortezomib exposure is increased in patients with moderate or severe hepatic impairment; these patients should be treated with VELCADE at reduced starting doses and closely monitored for toxicities (See **Posology and Method of Administration, 4.2** and **Pharmacokinetic Properties, 5.2**).

Posterior Reversible Encephalopathy Syndrome (PRES)

There have been reports of PRES in patients receiving VELCADE. PRES is a rare, reversible, neurological disorder which can present with seizure, hypertension, headache, lethargy, confusion, blindness, and other visual and neurological disturbances. Brain imaging, preferably MRI (Magnetic Resonance Imaging), is used to confirm the diagnosis. In patients developing PRES, discontinue VELCADE. The safety of reinitiating VELCADE therapy in patients previously experiencing PRES is not known.

4.5 Interaction with Other Medications and Other Forms of Interaction

In vitro and animal *ex vivo* studies indicate that bortezomib is a weak inhibitor of cytochrome P450 (CYP) isozymes 1A2, 2C9, 2C19, 2D6 and 3A4. Based on the limited contribution (7%) of CYP2D6 to the metabolism of bortezomib, the CYP2D6 poor metabolizer phenotype is not expected to affect the overall disposition of bortezomib.

A drug-drug interaction study assessing the effect of ketoconazole, a potent CYP3A4 inhibitor, on the pharmacokinetics of VELCADE, showed a bortezomib AUC mean increase of 35%, based on data from 12 patients. Therefore, patients should be closely monitored when given bortezomib in combination with potent CYP3A4-inhibitors (e.g., ketoconazole, ritonavir).

In a drug-drug interaction study assessing the effect of omeprazole, a potent inhibitor of CYP2C19, on the pharmacokinetics of VELCADE there was no significant effect on the pharmacokinetics of bortezomib, based on data from 17 patients.

A drug-drug interaction study assessing the effect of rifampicin, a potent CYP3A4 inducer, on the pharmacokinetics of VELCADE showed a mean bortezomib AUC reduction of 45% based on data

from 6 patients. The concomitant use of VELCADE with strong CYP3A4 inducers is therefore not recommended, as efficacy may be reduced. Examples of CYP3A4 inducers are rifampicin, carbamazepine, phenytoin, phenobarbital and St. John's Wort. In the same drug-drug interaction study, the effect of dexamethasone, a weaker CYP3A4 inducer was assessed. There was no significant effect on bortezomib pharmacokinetics based on data from 7 patients.

A drug-drug interaction study assessing the effect of melphalan-prednisone on VELCADE showed a 17% increase in mean bortezomib AUC based on data from 21 patients. This is not considered clinically relevant.

During clinical trials, hypoglycemia and hyperglycemia were reported in diabetic patients receiving oral hypoglycemics. Patients on oral antidiabetic agents receiving VELCADE treatment may require close monitoring of their blood glucose levels and adjustment of the dose of their antidiabetic medication.

Patients should be cautioned about the use of concomitant medications that may be associated with peripheral neuropathy (such as amiodarone, anti-virals, isoniazid, nitrofurantoin, or statins), or with a decrease in blood pressure.

Drug Laboratory Test Interactions

None known.

4.6 Pregnancy, Lactation and Fertility

Pregnancy

Women of childbearing potential should avoid becoming pregnant while being treated with VELCADE.

No placental transfer studies have been conducted with bortezomib. There are no adequate and well-controlled studies in pregnant women. If VELCADE is used during pregnancy, or if the patient becomes pregnant while receiving this drug, the patient should be apprised of the potential hazard to the fetus.

Patients should be advised to use effective contraceptive measures to prevent pregnancy and to avoid breast feeding during treatment with VELCADE.

See Nonclinical Safety Data, 5.3, for more information.

Lactation

It is not known whether bortezomib is excreted in human milk. Because many drugs are excreted in human milk and because of the potential for serious adverse reactions in nursing infants from VELCADE, women should be advised against breast feeding while being treated with VELCADE.

Fertility

Nonclinical fertility studies with bortezomib were not performed, but evaluation of reproductive tissues has been performed in the general toxicity studies. In the 6-month rat toxicity study,

degenerative effects in the ovary were observed at doses ≥ 0.3 mg/m² (one-fourth of the recommended clinical dose), and degenerative changes in the testes occurred at 1.2 mg/m². VELCADE could have a potential effect on either male or female fertility

4.7 Effects on Ability to Drive and Use Machines

VELCADE may cause tiredness, dizziness, fainting, or blurred vision. Patients should be advised not to drive or operate machinery if they experience these symptoms.

4.8 Undesirable Effects

Clinical Trials

Summary of Clinical Trials of VELCADE IV in Patients with Relapsed/Refractory Multiple Myeloma

The safety and efficacy of VELCADE were evaluated in 3 studies at the recommended dose of 1.3 mg/m². These included a phase 3 randomized, comparative study, versus dexamethasone of 669 patients with relapsed or refractory multiple myeloma who had received 1-3 prior lines of therapy (M34101-039); a phase 2 single arm, open-label, multicenter study of 202 patients who had received at least 2 prior therapies and demonstrated disease progression on their most recent therapy (M34100-025); and a phase 2 dose-response clinical study in relapsed multiple myeloma for patients who had progressed or relapsed on or after first line therapy with VELCADE 1.0 mg/m² or 1.3 mg/m² (M34100-024).

Table 9: VELCADE Adverse Drug Reactions in Phase 2 and Phase 3 Relapsed/Refractory Multiple Myeloma Studies

MedDRA System Organ Class Preferred Term	Study No.	
	M34101-039 (N = 331)	M34100-024/ M34100-025 (N = 228 ^a)
Blood and lymphatic system disorders		
Thrombocytopenia	115 (35%)	97 (43%)
Anemia	87 (26%)	74 (32%)
Neutropenia	62 (19%)	55 (24%)
Leucopenia	24 (7%)	15 (7%)
Lymphopenia	15 (5%)	11 (5%)
Pancytopenia	2 (< 1%)	6 (3%)
Febrile neutropenia	1 (< 1%)	1 (< 1%)
Cardiac disorders		
Arrhythmias	4 (1%)	2 (< 1%)
Tachycardia	9 (3%)	17 (7%)
Atrial fibrillation	6 (2%)	2 (< 1%)
Palpitations	5 (2%)	4 (2%)
Acute development or exacerbation of cardiac failure, including CHF	7 (2%)	8 (4%)
Pulmonary edema	6 (2%)	3 (1%)
Cardiogenic shock ^b	1 (< 1%)	-
New onset of decreased left ventricular ejection fraction	1 (< 1%)	-

	Study No.	
MedDRA System Organ Class	M34101-039	M34100-024/
Preferred Term	(N = 331)	M34100-025 (N = 228 ^a)
Atrial flutter	1 (< 1%)	-
Bradycardia	3 (< 1%)	1 (< 1%)
Ear & labyrinth disorders		
Hearing impairment	1 (< 1%)	1 (< 1%)
Eye disorders		
Blurred vision	9 (3%)	25 (11%)
Conjunctival infection and irritation	14 (4%)	7 (3%)
Gastrointestinal (GI) disorders		
Constipation	140 (42%)	97 (43%)
Diarrhea	190 (57%)	116 (51%)
Nausea	190 (57%)	145 (64%)
Vomiting	117 (35%)	82 (36%)
Gastrointestinal and abdominal pain, excluding oral and throat	80 (24%)	48 (21%)
Dyspepsia	32 (10%)	30 (13%)
Pharyngolaryngeal pain	25 (8%)	19 (8%)
Gastroesophageal reflux	10 (3%)	1 (< 1%)
Eructation	2 (< 1%)	4 (2%)
Abdominal distension	14 (4%)	13 (6%)
Stomatitis and mouth ulceration	24 (7%)	10 (4%)
Dysphagia	4 (1%)	5 (2%)
GI hemorrhage (upper and lower GI tract) ^b	7 (2%)	3 (1%)
Rectal hemorrhage (includes hemorrhagic diarrhea)	7 (2%)	3 (1%)
Tongue ulceration	2 (< 1%)	1 (< 1%)
Retching	3 (< 1%)	2 (< 1%)
Upper GI hemorrhage	1 (< 1%)	-
Hematemesis	1 (< 1%)	-
Oral mucosal petechiae	3 (< 1%)	-
Ileus paralytic	1 (< 1%)	2 (< 1%)
General disorders and administration site conditions		
Asthenic conditions	201 (61%)	149 (65%)
weakness	40 (12%)	44 (19%)
fatigue	140 (42%)	118 (52%)
lethargy	12 (4%)	9 (4%)
malaise	13 (4%)	22 (10%)
Pyrexia	116 (35%)	82 (36%)
Rigors	37 (11%)	27 (12%)
Edema of the lower limbs	35 (11%)	27 (12%)
Neuralgia	21 (6%)	5 (2%)
Chest pain	26 (8%)	16 (7%)
Injection site pain and irritation	1 (< 1%)	1 (< 1%)

	Study No.	
MedDRA System Organ Class	M34101-039	M34100-024/
Preferred Term	(N = 331)	M34100-025 (N = 228 ^a)
Injection site phlebitis	1 (< 1%)	1 (< 1%)
Hepatobiliary disorders		
Hyperbilirubinemia	1 (< 1%)	-
Abnormal liver function tests	3 (< 1%)	2 (< 1%)
Hepatitis	2 (< 1%) in study M34101-040 ^c	-
Immune system disorders		
Drug hypersensitivity	1 (< 1%)	1 (< 1%)
Infections and infestations		
Upper respiratory tract infection	26 (8%)	41 (18%)
Nasopharyngitis	45 (14%)	17 (7%)
Lower respiratory tract and lung infections	48 (15%)	29 (13%)
Pneumonia ^b	21 (6%)	23 (10%)
Herpes zoster (including multidermatomal or disseminated)	42 (13%)	26 (11%)
Herpes simplex	25 (8%)	13 (6%)
Bronchitis	26 (8%)	6 (3%)
Postherpetic neuralgia	4 (1%)	1 (< 1%)
Sinusitis	14 (4%)	15 (7%)
Pharyngitis	6 (2%)	2 (< 1%)
Oral candidiasis	6 (2%)	3 (1%)
Urinary tract infection	13 (4%)	14 (6%)
Catheter related infection	10 (3%)	6 (3%)
Sepsis and bacteremia ^b	9 (3%)	9 (4%)
Gastroenteritis	7 (2%)	-
Injury, poisoning, and procedural complications		
Catheter related complication	7 (2%)	8 (4%)
Investigations		
Increased ALT	3 (< 1%)	10 (4%)
Increased AST	5 (2%)	12 (5%)
Increased alkaline phosphatase	6 (2%)	8 (4%)
Increased GGT	1 (< 1%)	4 (2%)
Metabolism and nutritional disorders		
Decreased appetite and anorexia	112 (34%)	99 (43%)
Dehydration	24 (7%)	42 (18%)
Hyperglycemia	5 (2%)	16 (7%)
Hypoglycemia	7 (2%)	4 (2%)
Hyponatremia	8 (2%)	18 (8%)
Tumor lysis syndrome	2 (<1%) in study	-

	Study No.	
MedDRA System Organ Class Preferred Term	M34101-039 (N = 331)	M34100-024/ M34100-025 (N = 228 ^a)
	M34101-040 ^c	
Musculoskeletal and connective tissue disorders		
Pain in limb	50 (15%)	59 (26%)
Myalgia	39 (12%)	32 (14%)
Arthralgia	45 (14%)	60 (26%)
Nervous system disorders		
Peripheral neuropathy ^d	120 (36%)	84 (37%)
Paresthesia and dysesthesia	91 (27%)	53 (23%)
Dizziness, excluding vertigo	45 (14%)	48 (21%)
Headache	85 (26%)	63 (28%)
Dysgeusia	17 (5%)	29 (13%)
Polyneuropathy	9 (3%)	1 (< 1%)
Syncope	8 (2%)	17 (7%)
Convulsions	4 (1%)	-
Loss of consciousness	2 (< 1%)	-
Ageusia	2 (< 1%)	-
Psychiatric disorders		
Anxiety	31 (9%)	32 (14%)
Renal and urinary disorders		
Renal Impairment and Failure	21 (6%)	21 (9%)
Difficulty in micturition	2 (1%)	3 (1%)
Hematuria	5 (2%)	4 (2%)
Respiratory, thoracic, and mediastinal disorders		
Epistaxis	21 (6%)	23 (10%)
Cough	70 (21%)	39 (17%)
Dyspnea	65 (20%)	50 (22%)
Exertional dyspnea	21 (6%)	18 (8%)
Pleural effusion	4 (1%)	9 (4%)
Rhinorrhea	4 (1%)	14 (6%)
Hemoptysis	3 (< 1%)	2 (< 1%)
Skin and subcutaneous tissue disorders		
Skin rash, which can be pruritic, erythematous, and can include evidence of leukocytoclastic vasculitis	61 (18%)	47 (21%)
Urticaria	7 (2%)	5 (2%)
Vascular disorders		
Hypotension	20 (6%)	27 (12%)
Orthostatic/postural hypotension	14 (4%)	8 (4%)
Petechiae	6 (2%)	7 (3%)
Cerebral hemorrhage ^b	1 (< 1%)	-

^aAll 228 patients received VELCADE at a dose of 1.3 mg/m²

^bincludes fatal outcome

^cA study of VELCADE at the recommended dose of 1.3 mg/m² in multiple myeloma patients who experienced progressive disease after receiving at least four previous therapies or after receiving high-dose dexamethasone in Protocol M34101-039

^dincluding all preferred terms under the MedDRA HLT “peripheral neuropathy NEC”

Summary of Clinical Trials of VELCADE IV vs SC in Patients with Relapsed Multiple Myeloma

The safety and efficacy of VELCADE SC were evaluated in one Phase 3 study at the recommended dose of 1.3 mg/m². This was a randomized, comparative study of VELCADE IV vs SC in 222 patients with relapsed multiple myeloma.

Table 10: Incidence of VELCADE Adverse Drug Reactions reported in ≥ 10% of patients in the Phase 3 Relapsed Multiple Myeloma Study comparing VELCADE IV and SC

MedDRA System Organ Class	----- IV ----- (N=74)			----- SC ----- (N=147)		
	Total	---- Toxicity Grade, n (%)		Total	---- Toxicity Grade, n (%)	
	n (%)	3	≥ 4	n (%)	3	≥ 4
Blood and lymphatic system disorders						
Anemia	26 (35)	6 (8)	0	53 (36)	14 (10)	4 (3)
Leukopenia	16 (22)	4 (5)	1 (1)	29 (20)	9 (6)	0
Neutropenia	20 (27)	10 (14)	3 (4)	42 (29)	22 (15)	4 (3)
Thrombocytopenia	27 (36)	8 (11)	6 (8)	52 (35)	12 (8)	7 (5)
Gastrointestinal disorders						
Abdominal pain	8 (11)	0	0	5 (3)	1 (1)	0
Abdominal pain upper	8 (11)	0	0	3 (2)	0	0
Constipation	11 (15)	1 (1)	0	21 (14)	1 (1)	0
Diarrhea	27 (36)	3 (4)	1 (1)	35 (24)	2 (1)	1 (1)
Nausea	14 (19)	0	0	27 (18)	0	0
Vomiting	12 (16)	0	1 (1)	17 (12)	3 (2)	0
General disorders and administration site conditions						
Asthenia	14 (19)	4 (5)	0	23 (16)	3 (2)	0
Fatigue	15 (20)	3 (4)	0	17 (12)	3 (2)	0
Pyrexia	12 (16)	0	0	28 (19)	0	0
Infections and infestations						
Herpes zoster	7 (9)	1 (1)	0	16 (11)	2 (1)	0
Metabolism and nutrition disorders						
Decreased appetite	7 (9)	0	0	14 (10)	0	0
Musculoskeletal and connective tissue disorders						
Pain in extremity	8 (11)	2 (3)	0	8 (5)	1 (1)	0
Nervous system disorders						
Headache	8 (11)	0	0	5 (3)	0	0
Neuralgia	17 (23)	7 (9)	0	35 (24)	5 (3)	0
Peripheral sensory neuropathy	36 (49)	10 (14)	1 (1)	51 (35)	7 (5)	0

MedDRA System Organ Class	IV (N=74)			SC (N=147)		
	Total	Toxicity Grade, n (%)		Total	Toxicity Grade, n (%)	
	n (%)	3	≥ 4	n (%)	3	≥ 4
Psychiatric disorders						
Insomnia	8 (11)	0	0	18 (12)	0	0
Respiratory, thoracic and mediastinal disorders						
Dyspnea	9 (12)	2 (3)	0	11 (7)	2 (1)	0

Note: Percentages in 'Total' column for each group calculated with the number of subjects in each group as denominator.
Percentages of toxicity grade sub-groups calculated with the number of subjects in each group as denominator.

Although, in general safety data were similar for the IV and SC treatment groups, the following table highlights differences larger than 10% in the overall incidence of adverse drug reactions between the two treatment arms.

Table 11: Incidence of Adverse Drug Reactions with >10% Difference in Overall Incidence between Treatment Arms in the Phase 3 Relapsed Multiple Myeloma Study comparing VELCADE IV and SC, by Toxicity Grade and Discontinuation

MedDRA System Organ Class	IV (N=74)			SC (N=147)		
	Category, n (%)			Category, n (%)		
	TEAE	G ≥ 3	Disc	TEAE	G ≥ 3	Disc
MedDRA High Level Term						
All subjects with TEAE	73 (99)	52 (70)	20 (27)	140 (95)	84 (57)	33 (22)
Gastrointestinal disorders						
Diarrhea (excl infective)	27 (36)	4 (5)	1 (1)	35 (24)	3 (2)	1 (1)
Gastrointestinal and abdominal pains (excl oral and throat)	14 (19)	0	0	9 (6)	1 (1)	0
General disorders and administration site conditions						
Asthenic conditions	29 (39)	7 (9)	1 (1)	40 (27)	6 (4)	2 (1)
Infections and infestations						
Upper respiratory tract infections	19 (26)	2 (3)	0	20 (14)	0	0
Nervous system disorders						
Peripheral neuropathies ^a	39 (53)	12 (16)	10 (14)	56 (38)	9 (6)	9 (6)

^a Represents the high level term

TEAE = Treatment-Emergent Adverse Event; G ≥ 3 = Toxicity Grade greater than or equal to 3

Disc = Discontinuation of any study drug

Patients who received VELCADE subcutaneously compared to intravenous administration had 13% lower overall incidence of treatment-emergent adverse drug reactions that were grade 3 or higher in toxicity (57% vs 70% respectively), and a 5% lower incidence of discontinuation of VELCADE (22% vs 27%). The overall incidence of diarrhea (24% for the SC arm vs 36% for the

IV arm), gastrointestinal and abdominal pain (6% for the SC arm vs 19% for the IV arm), asthenic conditions (27% for SC arm vs 39% for IV arm), upper respiratory tract infections (14% SC arm vs 26% IV arm) and peripheral neuropathy NEC (38% SC arm vs 53% IV arm) were 12%-15% lower in the subcutaneous group than the intravenous group. In addition, the incidence of peripheral neuropathies that were grade 3 or higher in toxicity was 10% lower (6% for SC vs 16% for IV), and the discontinuation rate due to peripheral neuropathies was 8% lower for the subcutaneous group (5%) as compared to the intravenous group (12%).

Six percent of patients were reported to have had an adverse local reaction to SC administration, mostly redness. Only 2 (1%) subjects were reported as having severe reactions. These severe local reactions were 1 case of pruritus and 1 case of redness. These reactions seldom led to dose modifications and all resolved in a median of 6 days.

VELCADE Retreatment in Relapsed Multiple Myeloma

The following table describes adverse drug reactions reported for at least 10% of patients with relapsed multiple myeloma who received retreatment with VELCADE IV (Study MMY-2036).

Table 12: Incidence of VELCADE Adverse Drug Reactions reported in $\geq 10\%$ of patients (Study MMY-2036)

	Vc Retreatment (MMY-2036)		
	Total	3	≥ 4
Analysis Set: Safety, N	130		
Total no. subjects with adverse drug reactions, n (%)	126 (97)		
MedDRA	system	organ	class
Preferred term			
Blood and lymphatic system disorders			
Thrombocytopenia	71 (55)	19 (15)	14 (11)
Anemia	48 (37)	5 (4)	1 (1)
Neutropenia	23 (18)	9 (7)	0
Leukopenia	20 (15)	5 (4)	0
Gastrointestinal disorders			
Diarrhea	45 (35)	9 (7)	0
Constipation	36 (28)	0	0
Nausea	14 (11)	0	0
General disorders and administration site conditions			
Pyrexia	31 (24)	2 (2)	0
Asthenia	29 (22)	6 (5)	0
Fatigue	21 (16)	0	0
Edema peripheral	15 (12)	0	0
Infections and infestations			
Respiratory tract infection	17 (13)	3 (2)	1 (1)
Bronchitis	13 (10)	1 (1)	0
Nervous system disorders			
Peripheral sensory neuropathy	22 (17)	4 (3)	0
Neuropathy peripheral	13 (10)	3 (2)	0
Respiratory, thoracic and mediastinal disorders			
Cough	15 (12)	1 (1)	0
Dyspnea	14 (11)	1 (1)	0

	Vc Retreatment (MMY-2036)	
	Total	Toxicity Grade
		3 ≥4

Key: Vc = VELCADE; AE = Adverse event; NCI = National Cancer Institute; CTCAE = Common Toxicity Criteria for Adverse Events

Note: Percentages are calculated with the number of subjects in each group as denominator.

Adverse events are reported using MedDRA version 14.1.

In Study MMY-2036, for AEs where only a severity grade is reported, the severity grade is remapped to an NCI CTCAE toxicity grade.

AEs with missing toxicity grade are assigned grade 3.

Summary of Clinical Trials of VELCADE Combination therapy in Patients with Relapsed Multiple Myeloma

The following table describe adverse drug reactions reported for at least 10% of patients with relapsed multiple myeloma who received VELCADE in combination with dexamethasone (Study MMY-2045) or VELCADE in combination with pegylated liposomal doxorubicin (Study DOXIL-MMY-3001).

Table 13: Most Frequent (at Least 10 Percent in Any Treatment Group) Treatment-Emergent Adverse Drug Reactions by Toxicity Grade, System Organ Class and Preferred Term; Safety Analysis Set (Studies DOXIL-MMY-3001 and MMY-2045)

	Vc Combination Therapy						
	Vc Monotherapy		Vc + DOXIL			Vc + Dex	
	Total n (%)	Grade n (%)	≥3 n (%)	Total n (%)	Grade n (%)	≥3 n (%)	Total n (%)
Analysis Set: Safety	318		318			163	
Total no. subjects with adverse drug reactions	301 (95)		314 (99)			154 (94)	
MedDRA system organ class							
Preferred term							
Gastrointestinal disorders							
Diarrhea	124 (39)	16 (5)	145 (46)	23 (7)	51 (31)	7 (4)	
Nausea	126 (40)	3 (1)	154 (48)	8 (3)	20 (12)	1 (1)	
Constipation	98 (31)	2 (1)	99 (31)	3 (1)	50 (31)	9 (6)	
Vomiting	69 (22)	3 (1)	101 (32)	13 (4)	11 (7)	2 (1)	
Stomatitis	11 (3)	1 (< 1)	56 (18)	7 (2)	1 (1)	0	
Abdominal pain	24 (8)	4 (1)	34 (11)	2 (1)	11 (7)	1 (1)	
Nervous system disorders							
Peripheral neuropathy ^a	143 (45)	35 (11)	133 (42)	22 (7)	79 (48)	23 (14)	
Neuralgia	63 (20)	14 (4)	54 (17)	9 (3)	26 (16)	4 (2)	
Headache	56 (18)	0	59 (19)	3 (1)	9 (6)	0	
Paraesthesia	31 (10)	0	41 (13)	1 (< 1)	22 (13)	2 (1)	
Dizziness	26 (8)	4 (1)	32 (10)	4 (1)	14 (9)	0	
General disorders and administration site conditions							
Fatigue	88 (28)	8 (3)	115 (36)	22 (7)	37 (23)	2 (1)	
Pyrexia	71 (22)	4 (1)	100 (31)	4 (1)	21 (13)	4 (2)	
Asthenia	56 (18)	12 (4)	71 (22)	19 (6)	33 (20)	2 (1)	
Edema peripheral	27 (8)	1 (< 1)	32 (10)	1 (< 1)	43 (26)	3 (2)	
Blood and lymphatic system disorders							
Thrombocytopenia	89 (28)	53 (17)	106 (33)	76 (24)	61 (37)	28 (17)	
Neutropenia	71 (22)	51 (16)	114 (36)	102 (32)	12 (7)	6 (4)	
Anemia	68 (21)	30 (9)	80 (25)	29 (9)	35 (21)	16 (10)	
Infections and infestations							

	Vc Monotherapy		Vc Combination Therapy				
			Vc + DOXIL		Vc + Dex		
	Total n (%)	Grade n (%)	≥3 n (%)	Total n (%)	Grade n (%)	≥3 n (%)	Grade n (%)
Herpes zoster	29 (9)	6 (2)	34 (11)	6 (2)	16 (10)	1 (1)	
Bronchitis	21 (7)	3 (1)	31 (10)	1 (< 1)	18 (11)	1 (1)	
Upper respiratory tract infection	33 (10)	3 (1)	33 (10)	2 (1)	15 (9)	3 (2)	
Musculoskeletal and connective tissue disorders							
Back pain	39 (12)	6 (2)	39 (12)	4 (1)	25 (15)	2 (1)	
Pain in extremity	48 (15)	8 (3)	34 (11)	1 (< 1)	16 (10)	2 (1)	
Arthralgia	27 (8)	5 (2)	34 (11)	1 (< 1)	14 (9)	1 (1)	
Respiratory, thoracic and mediastinal disorders							
Cough	38 (12)	0	58 (18)	0	26 (16)	1 (1)	
Dyspnea	28 (9)	10 (3)	34 (11)	3 (1)	13 (8)	3 (2)	
Metabolism and nutritional disorders							
Decreased appetite	50 (16)	1 (< 1)	83 (26)	8 (3)	9 (6)	0	
Skin and subcutaneous tissue disorders							
Rash	29 (9)	3 (1)	48 (15)	2 (1)	8 (5)	0	
Investigations							
Weight decreased	12 (4)	0	37 (12)	0	3 (2)	0	
Psychiatric disorders							
Insomnia	43 (14)	2 (1)	35 (11)	0	18 (11)	1 (1)	

Key: Vc = VELCADE; Dex = dexamethasone; NCI = National Cancer Institute; CTCAE = Common Toxicity Criteria for Adverse Events

^a Includes the preferred terms of neuropathy peripheral, peripheral sensory neuropathy, peripheral motor neuropathy, peripheral sensorimotor neuropathy, and polyneuropathy.

Note: Percentages are calculated with the number of subjects in each group as denominator.

Adverse events are reported using MedDRA version 14.1.

In Study MMY-2045, for AEs where only a severity grade is reported, the severity grade is remapped to an NCI CTCAE toxicity grade.

Summary of Clinical Trials in Patients with Previously Untreated Multiple Myeloma

The following table describes safety data from 340 patients with previously untreated multiple myeloma who received VELCADE (1.3 mg/m²) in combination with melphalan (9 mg/m²) and prednisone (60 mg/m²) in a prospective phase 3 study.

Table 14: Treatment-Emergent Drug-Related Adverse Events Reported in ≥ 10% of patients treated with VELCADE IV in Combination with melphalan and prednisone

	Vc-MP			MP		
	(n=340)			(n=337)		
MedDRA System Organ Class	Total	Toxicity Grade, n (%)		Total	Toxicity Grade, n (%)	
Preferred Term	n (%)	3	≥4	n (%)	3	≥4
Blood and Lymphatic System Disorders						
Thrombocytopenia	164 (48)	60 (18)	57 (17)	140 (42)	48 (14)	39 (12)
Neutropenia	160 (47)	101 (30)	33 (10)	143 (42)	77 (23)	42 (12)

	Vc-MP			MP		
	(n=340)			(n=337)		
MedDRA System Organ Class	Total	Toxicity Grade, n (%)		Total	Toxicity Grade, n (%)	
Preferred Term	n (%)	3	≥4	n (%)	3	≥4
Anemia	109 (32)	41 (12)	4 (1)	156 (46)	61 (18)	18 (5)
Leukopenia	108 (32)	64 (19)	8 (2)	93 (28)	53 (16)	11 (3)
Lymphopenia	78 (23)	46 (14)	17 (5)	51 (15)	26 (8)	7 (2)
Gastrointestinal Disorders						
Nausea	134 (39)	10 (3)	0	70 (21)	1 (<1)	0
Diarrhea	119 (35)	19 (6)	2 (1)	20 (6)	1 (<1)	0
Vomiting	87 (26)	13 (4)	0	41 (12)	2 (1)	0
Constipation	77 (23)	2 (1)	0	14 (4)	0	0
Abdominal Pain Upper	34 (10)	1 (<1)	0	20 (6)	0	0
Nervous System Disorders						
Peripheral Neuropathy	156 (46)	42 (12)	2 (1)	4 (1)	0	0
Neuralgia	117 (34)	27 (8)	2 (1)	1 (<1)	0	0
Paraesthesia	42 (12)	6 (2)	0	4 (1)	0	0
General Disorders and Administration Site Conditions						
Fatigue	85 (25)	19 (6)	2 (1)	48 (14)	4 (1)	0
Asthenia	54 (16)	18 (5)	0	23 (7)	3 (1)	0
Pyrexia	53 (16)	4 (1)	0	19 (6)	1 (<1)	1 (<1)
Infections and Infestations						
Herpes Zoster	39 (11)	11 (3)	0	9 (3)	4 (1)	0
Metabolism and Nutrition Disorders						
Anorexia	64 (19)	6 (2)	0	19 (6)	0	0
Skin and Subcutaneous Tissue Disorders						
Rash	38 (11)	2 (1)	0	7 (2)	0	0
Psychiatric Disorders						
Insomnia	35 (10)	1 (<1)	0	21 (6)	0	0

Herpes Zoster Virus Reactivation:

Physician should consider using antiviral prophylaxis in patients being treated with VELCADE. In the phase 3 study in patients with previously untreated multiple myeloma, the overall incidence of herpes zoster reactivation was more common in patients treated with VcMP compared with MP (14% vs 4% respectively). Antiviral prophylaxis was administered to 26% of the patients in the VcMP arm. The incidence of herpes zoster among patients in the VcMP treatment group was 17% for patients not administered antiviral prophylaxis compared to 3% for patients administered antiviral prophylaxis.

Adverse Drug Reactions Noted in Patients with Previously Untreated Multiple Myeloma Eligible for Stem Cell Transplantation

The following table describes adverse drug reactions considered by the Company to have at least a possible causal relationship to VELCADE from patients with previously untreated multiple myeloma eligible for stem cell transplantation who received VELCADE IV (1.3 mg/m²). 410 patients were treated with VELCADE in combination with doxorubicin and dexamethasone compared with 411 patients treated with vincristine, doxorubicin and dexamethasone in Study MMY-3003, 239 were treated with VELCADE in combination with dexamethasone alone compared with 239 patients treated with vincristine, doxorubicin and dexamethasone in Study IFM

2005-01, and 130 were treated with VELCADE in combination with thalidomide and dexamethasone compared with 126 patients treated with thalidomide and dexamethasone in Study MMY-3010. For these 3 studies conducted in the transplant setting (MMY3003, IFM2005-01, MMY3010), only the adverse reactions during the induction phase of treatment are considered for the table.

Table 15: Incidence of Most Frequent ($\geq 10\%$ in Either Treatment Group) Treatment-Emergent Adverse Drug Reactions during Induction Stage (VELCADE Transplant Integrated Analysis of Safety: Safety Analysis Set)

MedDRA System Organ Class Preferred Term	----- Vc-Based -----			----- Non Vc-Based -----		
	Total n (%)	(N=779) - TOXICITY GRADE – n(%)		Total n (%)	(N=776) - TOXICITY GRADE – n (%)	
		2	≥ 3		2	≥ 3
Total no. subjects with adverse drug reactions	715 (92)			679 (88)		
Gastrointestinal disorders						
Constipation	242 (31)	89 (11)	10 (1)	214 (28)	67 (9)	8 (1)
Nausea	215 (28)	71 (9)	22 (3)	206 (27)	77 (10)	9 (1)
Diarrhea	133 (17)	29 (4)	23 (3)	110 (14)	26 (3)	6 (1)
Vomiting	95 (12)	30 (4)	18 (2)	87 (11)	35 (5)	6 (1)
Nervous system disorders						
Neuropathy peripheral	147 (19)	53 (7)	20 (3)	54 (7)	11 (1)	4 (1)
Paraesthesia	101 (13)	24 (3)	11 (1)	80 (10)	15 (2)	2 (<1)
Peripheral sensory neuropathy	101 (13)	41 (5)	19 (2)	55 (7)	13 (2)	1 (<1)
Headache	64 (8)	23 (3)	4 (1)	76 (10)	23 (3)	1 (<1)
General disorders and administration site conditions						
Fatigue	158 (20)	50 (6)	21 (3)	161 (21)	68 (9)	21 (3)
Pyrexia	153 (20)	56 (7)	25 (3)	159 (20)	40 (5)	36 (5)
Asthenia	110 (14)	33 (4)	16 (2)	91 (12)	33 (4)	10 (1)
Blood and lymphatic system disorders						
Thrombocytopenia	239 (31)	54 (7)	63 (8)	171 (22)	27 (3)	27 (3)
Anemia	211 (27)	95 (12)	55 (7)	222 (29)	108 (14)	77 (10)
Leukopenia	196 (25)	51 (7)	109 (14)	206 (27)	53 (7)	120 (15)
Infections and infestations						
Herpes zoster	86 (11)	50 (6)	24 (3)	18 (2)	9 (1)	5 (1)
Metabolism and nutrition disorders						
Hyperglycemia	122 (16)	46 (6)	26 (3)	138 (18)	46 (6)	31 (4)
Hyponatremia	100 (13)	2 (<1)	29 (4)	82 (11)	6 (1)	12 (2)
Psychiatric disorders						
Insomnia	96 (12)	32 (4)	6 (1)	82 (11)	30 (4)	6 (1)

Vc=VELCADE

Note: Percentages for each group calculated with the number of subjects in each group as denominator. Incidence is based on the number of subjects experiencing at least 1 adverse reaction, not the number of events.

Adverse Events are coded using MedDRA 13.1.

Summary of the Clinical Trial in Patients with Relapsed Mantle Cell Lymphoma

Safety data for patients with relapsed mantle cell lymphoma were evaluated in a phase 2 study [M34103-053 (PINNACLE)], which included 155 patients treated with VELCADE at the recommended dose of 1.3 mg/m². The safety profile of VELCADE in these patients was similar to that observed in patients with multiple myeloma. Notable differences between the two patient populations were that thrombocytopenia, neutropenia, anemia, nausea, vomiting and pyrexia were reported more often in the patients with multiple myeloma than in those with mantle cell lymphoma; whereas peripheral neuropathy, rash and pruritus were higher among patients with mantle cell lymphoma compared to patients with multiple myeloma.

Summary of Clinical Trial in Patients with Previously Untreated Mantle Cell Lymphoma

Table 16 describes safety data from 240 patients with previously untreated mantle cell lymphoma who received VELCADE (1.3 mg/m²) administered IV in combination with rituximab (375 mg/m²), cyclophosphamide (750 mg/m²), doxorubicin (50 mg/m²), and prednisone (100 mg/m²) (VcR-CAP) in a prospective randomized study.

The incidences of Grade ≥ 3 bleeding events were similar between the 2 arms (4 patients in the VcR-CAP arm and 3 patients in the R-CHOP arm). All of the Grade ≥ 3 bleeding events resolved without sequelae in the VcR-CAP arm.

Infections were reported for 31% of patients in the VcR-CAP arm and 23% of the patients in the R-CHOP arm. Respiratory tract and lung infections were reported, with the predominant preferred term of pneumonia (VcR-CAP 8% versus R-CHOP 5%).

The incidence of herpes zoster reactivation was 4.6% in the VcR-CAP arm and 0.8% in the R-CHOP arm. Antiviral prophylaxis was mandated by protocol amendment.

Table 16: Most Commonly Reported Adverse Reactions ($\geq 5\%$) with Grades 3 and ≥ 4 Intensity in the Mantle Cell Lymphoma Study of VcR-CAP versus R-CHOP (N=482) (Study LYM-3002)

		VcR-CAP n=240		R-CHOP n=242		
System Organ Class	Total	Toxicity	Toxicity	Total	Toxicity	Toxicity
Preferred Term	n (%)	Grade 3	Grade ≥ 4	n (%)	Grade 3	Grade ≥ 4
		n (%)	n (%)		n (%)	n (%)
Blood and lymphatic system disorders						
Neutropenia	209 (87)	32 (13)	168 (70)	172 (71)	31 (13)	125 (52)
Leukopenia	116 (48)	34 (14)	69 (29)	87 (36)	39 (16)	27 (11)
Anemia	106 (44)	27 (11)	4 (2)	71 (29)	23 (10)	4 (2)
Thrombocytopenia	172 (72)	59 (25)	76 (32)	42 (17)	9 (4)	3 (1)
Febrile neutropenia	41 (17)	24 (10)	12 (5)	33 (14)	17 (7)	15 (6)

		VcR-CAP n=240		R-CHOP n=242		
System Organ Class	Total	Toxicity Grade 3	Toxicity Grade ≥4	Total	Toxicity Grade 3	Toxicity Grade ≥4
Preferred Term	n (%)	n (%)	n (%)	n (%)	n (%)	n (%)
Lymphopenia	68 (28)	25 (10)	36 (15)	28 (12)	15 (6)	2 (1)
Nervous system disorders						
Peripheral sensory neuropathy	53 (22)	11 (5)	1 (< 1)	45 (19)	6 (3)	0
Neuropathy peripheral	18 (8)	4 (2)	0	18 (7)	2 (1)	0
Hypoesthesia	14 (6)	3 (1)	0	13 (5)	0	0
Paraesthesia	14 (6)	2 (1)	0	11 (5)	0	0
Neuralgia	25 (10)	9 (4)	0	1 (< 1)	0	0
General disorders and administration site conditions						
Fatigue	43 (18)	11 (5)	1 (< 1)	38 (16)	5 (2)	0
Pyrexia	48 (20)	7 (3)	0	23 (10)	5 (2)	0
Asthenia	29 (12)	4 (2)	1 (< 1)	18 (7)	1 (< 1)	0
Edema peripheral	16 (7)	1 (< 1)	0	13 (5)	0	0
Gastrointestinal disorders						
Nausea	54 (23)	1 (< 1)	0	28 (12)	0	0
Constipation	42 (18)	1 (< 1)	0	22 (9)	2 (1)	0
Stomatitis	20 (8)	2 (1)	0	19 (8)	0	1 (< 1)
Diarrhea	59 (25)	11 (5)	0	11 (5)	3 (1)	1 (< 1)
Vomiting	24 (10)	1 (< 1)	0	8 (3)	0	0
Abdominal distension	13 (5)	0	0	4 (2)	0	0
Infections and infestations						
Pneumonia	20 (8)	8 (3)	5 (2)	11 (5)	5 (2)	3 (1)
Skin and subcutaneous tissue disorders						
Alopecia	31 (13)	1 (< 1)	1 (< 1)	33 (14)	4 (2)	0
Metabolism and nutrition disorders						
Hyperglycemia	10 (4)	1 (< 1)	0	17 (7)	10 (4)	0
Decreased appetite	36 (15)	2 (1)	0	15 (6)	1 (< 1)	0
Hypokalemia	11 (5)	3 (1)	1 (< 1)	6 (2)	1 (< 1)	0
Vascular disorders						
Hypertension	15 (6)	1 (< 1)	0	3 (1)	0	0
Psychiatric disorders						
Insomnia	16 (7)	1 (< 1)	0	8 (3)	0	0

Key: R-CHOP=rituximab, cyclophosphamide, doxorubicin, vincristine, and prednisone; VcR-CAP=VELCADE, rituximab, cyclophosphamide, doxorubicin, and prednisone.

Postmarketing

Following is a list of ADRs which have been observed in postmarketing and are not included above:

Clinically significant adverse drug reactions are listed here if they have not been reported above.

The frequencies provided below reflect reporting rates of adverse drug reactions from the worldwide post-marketing experience with VELCADE. The frequencies provided below reflect reporting rates and precise estimates of incidence cannot be made. The following convention is used for the classification of the frequency of an adverse drug reaction (ADR) and is based on the Council for International Organizations of Medical Sciences (CIOMS) guidelines: very common

($\geq 1/10$), common ($\geq 1/100$ and $< 1/10$), uncommon ($\geq 1/1,000$ and $< 1/100$), rare ($\geq 1/10,000$ and $< 1/1,000$), very rare ($< 1/10,000$).

Table 17: Post-Marketing Reports of Adverse Reactions

Blood and lymphatic system disorders	
Disseminated intravascular coagulation	Rare
Thrombotic microangiopathy	Very Rare
Cardiac disorders	
Atrioventricular block complete, cardiac tamponade	Rare
Ear and labyrinth disorders	
Deafness bilateral	Rare
Eye disorders	
Ophthalmic herpes, optic neuropathy, blindness	Rare
Chalazion/blepharitis	Rare
Gastrointestinal disorders	
Ischemic colitis, acute pancreatitis	Rare
Intestinal obstruction	Uncommon
Infections and infestations	
Herpes meningoencephalitis, septic shock	Rare
Progressive multifocal leukoencephalopathy ^a	Very rare
Immune system disorders	
Angioedema	Rare
Anaphylactic reaction	Very rare
Nervous system disorders	
Encephalopathy, autonomic neuropathy, posterior reversible encephalopathy syndrome	Rare
Guillain-Barré syndrome, demyelinating polyneuropathy	Rare
Respiratory, thoracic and mediastinal disorders	
Acute diffuse infiltrative pulmonary disease (see Special Warnings and Special Precautions for Use, 4.4)	Rare Rare
Pulmonary hypertension	
Skin and subcutaneous tissue disorders	
Stevens-Johnson Syndrome and toxic epidermal necrolysis	Very rare
Acute febrile neutrophilic dermatosis (Sweet's syndrome)	Rare

^a Very rare cases with unknown causality of John Cunningham (JC) virus infection, resulting in PML and death, have been reported in patients treated with VELCADE.

4.9 Overdose

Signs and Symptoms

Cardiovascular safety pharmacology studies in monkeys and dogs show that IV doses approximately two to three times the recommended clinical dose on a mg/m² basis are associated

with increases in heart rate, decreases in contractility, hypotension and death. The decreased cardiac contractility and hypotension responded to acute intervention with positive inotropic or pressor agents. In dog studies, a slight increase in the corrected QT interval was observed at a lethal dose.

Overdosage more than twice the recommended dose in patients has been associated with the acute onset of symptomatic hypotension and thrombocytopenia with fatal outcomes.

Management

There is no known specific antidote for VELCADE overdosage. In the event of an overdosage, the patient's vital signs should be monitored and appropriate supportive care given to maintain blood pressure (such as fluids, pressors, and/or inotropic agents) and body temperature (see **Special Warnings and Special Precautions for Use, 4.4** and **Posology and Method of Administration, 4.2**).

4.10 Drug Abuse and Dependence

Bortezomib has no known potential for abuse or dependence.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic Properties

Mechanism of Action

Bortezomib is a reversible inhibitor of the chymotrypsin-like activity of the 26S proteasome in mammalian cells. The 26S proteasome is a large protein complex that degrades ubiquitinated proteins. The ubiquitin-proteasome pathway plays an essential role in regulating the intracellular concentration of specific proteins, thereby maintaining homeostasis within cells. Inhibition of the 26S proteasome prevents this targeted proteolysis which can affect multiple signaling cascades within the cell. This disruption of normal homeostatic mechanisms can lead to cell death. Experiments have demonstrated that bortezomib is cytotoxic to a variety of cancer cell types *in vitro*. Bortezomib causes a delay in tumor growth *in vivo* in nonclinical tumor models, including multiple myeloma.

5.2 Pharmacokinetic Properties

Following intravenous bolus administration of a 1.0 mg/m² and 1.3 mg/m² dose to eleven patients with multiple myeloma, the mean first-dose maximum plasma concentrations of bortezomib were 57 and 112 ng/mL, respectively. In subsequent doses, mean maximum observed plasma concentrations ranged from 67 to 106 ng/mL for the 1.0 mg/m² dose and 89 to 120 ng/mL for the 1.3 mg/m² dose.

In the PK/PD substudy in Phase 3 trial, following an IV bolus or subcutaneous (SC) injection of a 1.3 mg/m² dose to multiple myeloma patients (n = 14 for IV, n = 17 for SC), the total systemic exposure after repeat dose administration (AUC_{last}) was equivalent for SC and IV administration. The C_{max} after SC administration (20.4 ng/mL) was lower than IV (223 ng/mL). The AUC_{last} geometric mean ratio was 0.99 and 90% confidence intervals were 80.18% - 122.80%.

Distribution

The mean distribution volume of bortezomib ranged from 1659 liters to 3294 liters single- or repeat

dose IV administration of 1.0 mg/m² or 1.3 mg/m² to patients with multiple myeloma. This suggests that bortezomib distributes widely to peripheral tissues. The binding of bortezomib to human plasma proteins averaged 83% over the concentration range of 100 - 1000 ng/mL.

Metabolism

In vitro studies with human liver microsomes and human cDNA-expressed cytochrome P450 isozymes indicate that bortezomib is primarily oxidatively metabolized via cytochrome P450 enzymes, 3A4, 2C19, and 1A2. Bortezomib metabolism by CYP 2D6 and 2C9 enzymes is minor. The major metabolic pathway is deboronation to form two deboronated metabolites that subsequently undergo hydroxylation to several metabolites. Deboronated-bortezomib metabolites are inactive as 26S proteasome inhibitors. Pooled plasma data from 8 patients at 10 min and 30 min after dosing indicate that the plasma levels of metabolites are low compared to the parent drug.

Elimination

The mean elimination half-life of bortezomib upon multiple dosing ranged from 40-193 hours. Bortezomib is eliminated more rapidly following the first dose compared to subsequent doses. Mean total body clearances were 102 and 112 L/h following the first dose for doses of 1.0mg/m² and 1.3mg/m², respectively, and ranged from 15 to 32 L/h following subsequent doses for doses of 1.0mg/m² and 1.3mg/m², respectively.

The pathways of elimination of bortezomib have not been characterized in humans.

Special Populations

Impaired Renal Function

A pharmacokinetic study was conducted in patients with various degrees of renal impairment who were classified according to their creatinine clearance values (CrCL) into the following groups: Normal (CrCL ≥60 mL/min/1.73 m², n=12), Mild (CrCL=40-59 mL/min/1.73 m², n=10), Moderate (CrCL=20-39 mL/min/1.73 m², n=9), and Severe (CrCL < 20 mL/min/1.73 m², n=3). A group of dialysis patients who were dosed after dialysis was also included in the study (n=8). Patients were administered intravenous doses of 0.7 to 1.3 mg/m² of bortezomib twice weekly. Exposure of bortezomib (dose-normalized AUC and C_{max}) was comparable among all the groups (see **Posology and Method of Administration, 4.2**).

Impaired Hepatic Function

The effect of hepatic impairment on the pharmacokinetics of IV bortezomib was assessed in 60 cancer patients at bortezomib doses ranging from 0.5 to 1.3 mg/m². When compared to patients with normal hepatic function, mild hepatic impairment did not alter dose-normalized bortezomib AUC. However, the dose-normalized mean AUC values were increased by approximately 60% in patients with moderate or severe hepatic impairment. A lower starting dose is recommended in patients with moderate or severe hepatic impairment, and those patients should be monitored closely (see **Table 7**).

Age, Gender, Race

The pharmacokinetics of bortezomib were characterized following twice weekly intravenous bolus administration of 1.3 mg/m² doses to 104 pediatric patients (2-16 years old) with acute lymphoblastic leukemia (ALL) or acute myeloid leukemia (AML). Based on a population

pharmacokinetic analysis, clearance of bortezomib increased with increasing body surface area (BSA). Geometric mean (%CV) clearance was 7.79 (25%) L/hr/m², volume of distribution at steady-state was 834 (39%) L/m², and the elimination half-life was 100 (44%) hours. After correcting for the BSA effect, other demographics such as age, body weight and sex did not have clinically significant effects on bortezomib clearance. BSA-normalized clearance of bortezomib in pediatric patients was similar to that observed in adults.

The effects of gender, and race on the pharmacokinetics of bortezomib have not been evaluated.

Drug Interactions

See section 4.5 for relevant information.

5.3 Nonclinical Safety Data

Carcinogenesis, Mutagenesis, Reproductive Toxicology

Carcinogenicity studies have not been conducted with bortezomib.

Bortezomib showed clastogenic activity (structural chromosomal aberrations) in the *in vitro* chromosomal aberration assay using Chinese hamster ovary cells. Bortezomib was not genotoxic when tested in the *in vitro* mutagenicity assay (Ames test) and *in vivo* micronucleus assay in mice.

Bortezomib was not teratogenic in nonclinical developmental toxicity studies in rats and rabbits at the highest dose tested [0.075 mg/kg (0.5 mg/m²) in the rat and 0.05 mg/kg (0.6 mg/m²) in the rabbit] when administered during organogenesis. These dosages are approximately half the clinical dose of 1.3 mg/m² based on body surface area.

Pregnant rabbits given bortezomib during organogenesis at a dose of 0.05 mg/kg (0.6 mg/m²) experienced significant post-implantation loss and decreased number of live fetuses. Live fetuses from these litters also showed significant decreases in fetal weight. The dose is approximately 0.5 times the clinical dose of 1.3 mg/m² based on body surface area.

6. PHARMACEUTICAL PARTICULARS

6.1 List of Excipients

Mannitol (E421)

6.2 Incompatibilities

This product must not be mixed with other medicinal products except those mentioned in **Instructions for Use/Handling**.

6.3 Shelf Life

36 months

Unopened vials of VELCADE are stable until the date indicated on the package when stored in the original package protected from light.

6.4 Special Precautions for Storage

VELCADE contains no antimicrobial preservative. When reconstituted as directed, VELCADE may be stored at 25°C. Reconstituted VELCADE should be administered within 8 hours of preparation. The reconstituted material may be stored for up to 8 hours in the original vial or in a syringe. The total storage time for the reconstituted material must not exceed 8 hours when exposed to normal indoor lighting.

Do not store unopened vials above 30°C. Retain in original package to protect from light.

Keep out of reach of children.

6.5 Nature and Contents of Container

Five (5) or ten (10) mL, type 1, glass vial with a gray bromobutyl stopper and aluminum seal. The cap color of the 5 mL vial is green, and the cap for the 10 mL vial is royal blue for the 3.5 mg product presentation. Each vial is contained in a transparent blister pack consisting of a tray with a lid. The 5 mL vial contains 11 mg powder for solution for injection and the 10 mL vial contains 38.5 mg powder for solution for injection for the 3.5mg product.

VELCADE is available in cartons containing 1 single use vial. Not all pack size may be marketed.

6.6 Instructions for Use/Handling

Administration Precautions

VELCADE is an antineoplastic. Caution should be used during handling and preparation. Proper aseptic technique should be used. Use of gloves and other protective clothing to prevent skin contact is recommended. In clinical trials, local skin irritation was reported in 5% of patients, but extravasation of VELCADE was not associated with tissue damage.

When administered subcutaneously, alternate sites for each injection (thigh or abdomen). New injections should be given at least one inch from an old site and never into areas where the site is tender, bruised, red, or hard.

There have been fatal cases of inadvertent intrathecal administration of VELCADE. VELCADE is for IV and subcutaneous use only. **DO NOT ADMINISTER VELCADE INTRATHECALLY.**

Reconstitution/Preparation for Intravenous and Subcutaneous Administration

The contents of each vial should be reconstituted only with normal (0.9%) saline according to the following instructions based on route of administration:

	IV	SC
	3.5 mg bortezomib	3.5 mg bortezomib
Volume of diluent (0.9% Sodium Chloride) added to reconstitute one vial	3.5 mL	1.4 mL
Final Concentration after reconstitution (mg/mL)	1.0 mg/mL	2.5 mg/mL

The reconstituted product should be a clear and colorless solution. Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration whenever solution and container permit. If any discoloration or particulate matter is observed, the reconstituted product should not be used.

Procedure for Proper Disposal

Any unused product or waste material should be disposed of in accordance with local requirements.

PRODUCT REGISTRATION HOLDER

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